

Mateusz Płoszaj-Mazurek

Architect, PhD · Building researcher and computational designer

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(<https://github.com/Curiosit>)

Architect with ten years of practice in Denmark and Poland, a PhD in digital design methods and two years building design software for architects. I research daylight and building regulation at Warsaw University of Technology and build computational tools that make building performance part of design, not a check at the end.

Experience

current	Researcher and lecturer Faculty of Architecture, Warsaw University of Technology - Daylight regulation, simulation and surrogate-model research; see publications. - Teaches Grasshopper and multi-criteria building optimization; coordinates a lecture cycle on public buildings; runs a second-year design studio.
2024–2026	Software engineer, interim product owner One Click LCA, Helsinki · remote - Built features for Carbon Designer 3D, an early-design tool for architects (TypeScript, React, Three.js, AG Grid). - Interim product owner in the second half of 2025; took the Free Drawing feature from specification to release. - Wrote methodology specifications and brought in a paying customer.
2024	Contract developer enxess, Liechtenstein Web viewer for IFC models of historical building objects (Three.js, React, TypeScript).
2023–2024	Founder SLAD.AI Self-funded venture building early-design carbon estimation software; designed, built and tested with users.
2014–2024	Architect, Certified Passive House Designer Bjerg Arkitektur · Hjørring 2014–16, Warsaw 2016–24 - Single- and multi-family housing, schools, kindergartens and offices, from 100 m² to 15,000 m². - All design phases: concept, permit and construction documentation, and architectural supervision on site. - Code compliance in practice under Danish and Polish regulations and Passive House certification; daylight simulation studies with VELUX. 2022–2024: architecture, sustainability and software consultant, building in-house design tools; co-author of the Polish whole-life carbon roadmap.

Selected projects

2025–26	OmegaFlux Per-point daylight prediction from geometric view descriptors with gradient-boosted trees, exported to ONNX and run in the browser. Dataset generation, held-out validation against Radiance, transfer to unseen room shapes. curiosit.github.io/OmegaFlux (https://curiosit.github.io/OmegaFlux/)
2025	Free Drawing, Carbon Designer 3D One Click LCA Led a feature letting architects sketch building massing directly in the browser, from user problem through specification to release.
2021	Whole Life Carbon Roadmap for Poland PLGBC, WorldGBC, EBRD Co-author, with Alicja Kuczera, of the national roadmap for decarbonising the built environment by 2050.
2023–24	AEC Hackathons Copenhagen 2023 (SpeckleLCA), Zurich 2024 (InfraWeb), Copenhagen 2024 (OpenDetail, Best Mashup Project).

Publications	A workflow for LLM-assisted IFC data enrichment to support circular reuse of large-panel building systems in Poland with A. M. Tofiluk · Journal of Building Engineering 120, 115397 (2026) · doi · open access	
	Artificial Intelligence and Digital Tools for Assisting Low-Carbon Architectural Design: Merging the Use of Machine Learning, Large Language Models, and Building Information Modeling for Life Cycle Assessment Tool Development with E. Ryńska · Energies 17(12), 2997 (2024) · doi	
	Methods to Optimize Carbon Footprint of Buildings in Regenerative Architectural Design with the Use of Machine Learning, Convolutional Neural Network, and Parametric Design with E. Ryńska and M. Grochulska-Salak · Energies 13(20), 5289 (2020) · doi 136 citations	
	Analytic radiometric descriptors enable geometry-flexible per-sensor daylight prediction without learned spatial encodings (preprint) SSRN (2026) · code github.com/Curiosi/OmegaFlux · software doi:10.5281/zenodo.21781259	
	Akademik na Solcu, Warsaw Project publication, with M. Czechowicz · Architektura-Murator 4/2025, pp. 54–64	
	Przedszkole pasywne, Reguły Passive House kindergarten, project publication, with D. Sibińska, J. Sokołowski et al. · Architektura-Murator 5/2023, pp. 20–34	
Skills	Methods	Evolutionary and multi-objective optimization, surrogate modelling, gradient-boosted trees, dataset and validation design, parametric experiments
	Building domain	Climate-based daylight (sDA, ASE, UDI, EN 17037), insolation, Polish and Danish building regulations, Passive House, building carbon footprint
	Design tools	Rhino and Grasshopper, Ladybug and Honeybee, Radiance, THERM, IFC and BIM, That Open Company tools
	Code	Python (NumPy, pandas, scikit-learn, XGBoost, ONNX), TypeScript, React, Three.js, Git, CI/CD
	Languages	Polish (native), English (fluent)
Education	2022	PhD in Architecture Warsaw University of Technology Digital methods of supporting the architectural design process and the analysis of the carbon footprint of buildings: parametric optimization and machine-learning methods for design.
	2014–2016	MSc in Architecture Aalborg University, Denmark
	2023–2024	Certifications Master BIM Software Developer (That Open Company, 2024) · Harvard CS50x (2023) · Certified Passive House Designer
Grants and awards	2024–25	Principal investigator, IDUB research grant · Circular economy in architecture: reuse of prefabricated elements and digital design techniques · Warsaw University of Technology, CPR-IDUB/238/Z01/2024
	2024	First prize, Best Mashup Project · AEC Hackathon, Copenhagen
	2023	Competition win: energy-efficient multi-storey housing in Poland · Winning entry by Studio Widoki, with Bjerg Arkitektur as consultant · Carbon footprint calculations by Bjerg Arkitektur: prefabricated timber, about 6 kg CO ₂ e/m ² per year over 50 years
	2022	PLGBC Green Building Awards: best green building without certification · Passive House kindergarten in Podkowa Leśna, Bjerg Arkitektur; award received on behalf of the practice. Also a distinction for an educational building.
	2022	Distinction for the doctoral thesis · Innowator Mazowska a competition
Teaching	7 years	Introduction to Grasshopper Parametric modelling and multi-criteria optimization of energy, daylight and carbon.
Talks and panels	2023	Pro-environmental design: training sessions for architects · Saint-Gobain · two sessions, Jura Krakowsko-Częstochowska and Żnin, October 2023